

Power Electronics

BRANCH: ELECTRICAL (EE)

COURSE CODE: EE401

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Power Electronics (EE401) under the Electrical (EE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Power Semiconductor Devices

Characteristics of Power Diodes, SCR, Triac, MOSFET, IGBT. Thyristor triggering and commutation methods.

Unit 2: AC to DC Converters (Rectifiers)

Single-phase and Three-phase half/fully controlled bridge rectifiers with R, RL loads. Dual converters.

Unit 3: DC-DC Choppers

Step-down (Buck), Step-up (Boost), Buck-Boost choppers. Control strategies: PWM, frequency modulation.

Unit 4: Inverters & AC Controllers

Single-phase and Three-phase voltage source inverters (VSI). PWM techniques. Single-phase AC voltage controllers.

Average output voltage (Buck): $V_o = D \cdot V_i$. Thyristor average current rating calculation.

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers



